1 FIG-205 — Dft phase3 input audit

Group: DFT-PHASE3-INPUT-AUDIT **Evidence chain:** manifested **Status:** input audit only; no accepted target DFT output

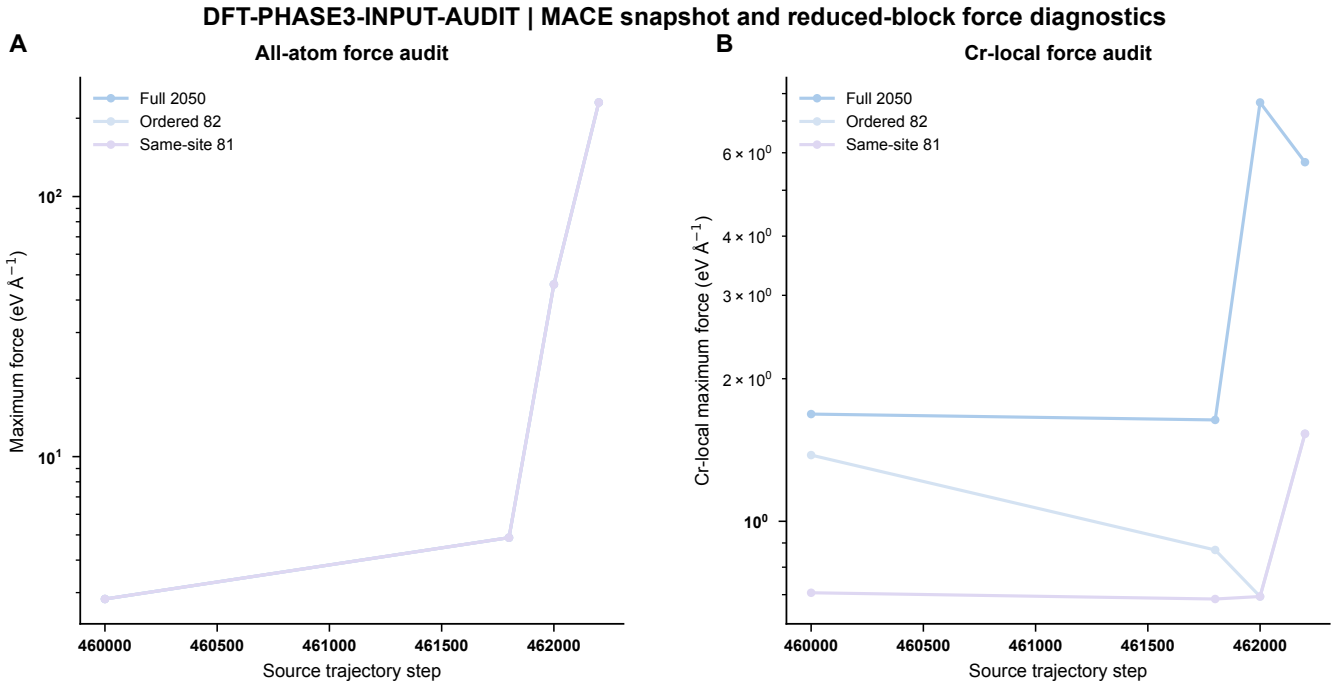


Figure 1: A, maximum all-atom force; B, Cr-local maximum force for the full 2,050-atom source snapshot and two reduced ordered-block representations across four source steps. This page shows the Plot Atlas table-backed candidate "Dft phase3 input audit"; its scientific status and caveats are reproduced from the package manifest.

Results text draft. The full source snapshot rises from 2.83 eV Å-1 at step 460000 to 229.97 eV Å-1 at step 462200. The reduced 82- and 81-atom representations retain the same source-step progression but are different validation units. **Discussion and boundary.** This is a pre-DFT input-integrity and stress-test figure. It documents why steps 462000 and 462200 are OOD/stress geometries and why a successful reduced-cell QE single point would still be a local comparison, not a converged migration barrier. **Article placement.** Proposed article location: Results/SI: reduced-cell DFT validation boundary. **Reviewer linkage:** overall status PARTIAL.

- **REV-A1 (PARTIAL)**: Reconcile Fig. 3d fixed-geometry DFT-path scoring with SI Fig. S3 self-consistent MLFF-NEB, including pathway identity and the DFT reference. *Completed:* The fixed-path versus self-relaxed operator distinction and the narrower model-reported-barrier wording are integrated. *Still required:* Close the restartable QE path-1–4 convergence gate, preserve the final path/source metadata, and complete the SI Fig. S3 endpoint/checkpoint/optimizer/source audit before calling the quantitative reference final.

Evidence pointers.

- Data/processed: main/analysis/dft_validation_phase3_20261002/snapshot_audit.csv
- Data/processed: main/analysis/dft_validation_phase3_20261002/ordered_block_audit.csv
- Plot/processing script: main/analysis/figure_evidence_audit_20261008/scripts/render_table_backed_packages.py
- Caption/text source: main/analysis/figure_evidence_audit_20261008/generated/table_backed_packages/DFT-PHASE3-INPUT-AUDIT/caption_results_discussion.md

Machine-collected evidence candidates (generated from the audit matrix; do not edit this file).

- **Caption/discussion source:** /Users/alina/Documents/ChatGPT/Wiley-MigrationBench/analysis/figure_evidence_audit_20261008/generated/table_backed_packages/DFT-PHASE3-INPUT-AUDIT/caption_results_discussion.md

- **Raw or processed data:** /Users/alina/Documents/ChatGPT/Wiley-MigrationBench/analysis/dft_validation_phase3_20261002/snapshot_audit.csv
- **Raw or processed data:** /Users/alina/Documents/ChatGPT/Wiley-MigrationBench/analysis/dft_validation_phase3_20261002/ordered_block_audit.csv
- **Raw or processed data:** /Users/alina/Documents/ChatGPT/Wiley-MigrationBench/analysis/figure_evidence_audit_20261008/generated/table_backed_packages/DFT-PHASE3-INPUT-AUDIT/processed_phase3_input_force_audit.csv
- **Plot or processing script:** /Users/alina/Documents/ChatGPT/Wiley-MigrationBench/analysis/figure_evidence_audit_20261008/scripts/render_table_backed_packages.py
- **Plot or processing script:** /Users/alina/Documents/Codex/plot_library

Audit gap / next action: These are MACE forces stored in the source extxyz audits, not QE forces. The 81-atom model removes one Cr and is explicitly a same-site composition approximation. No plot panel may be used to claim DFT convergence or a barrier.